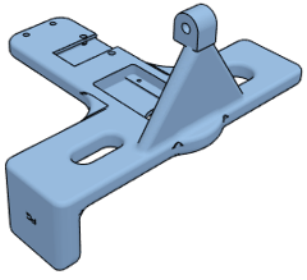
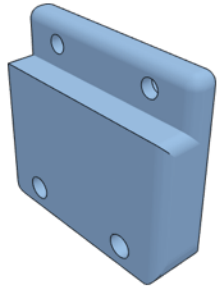
Parts

Print these (hardware/STEP or STL, right/ for the right hand, left/ for the left).



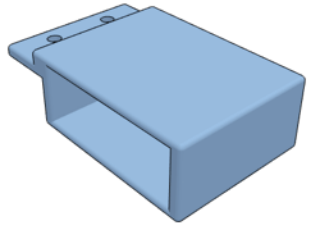
main_support

1×



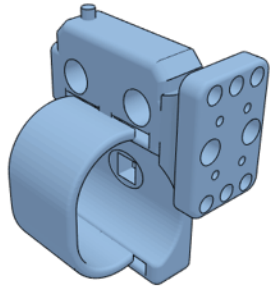
main_support_cover_plate

1×



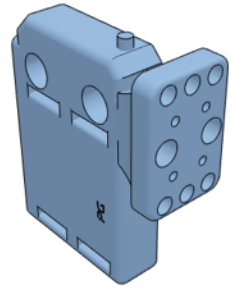
hand_support_base

1×



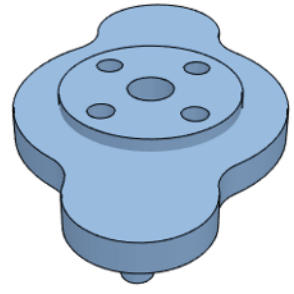
index_link

1×



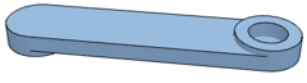
thumb_link

1×



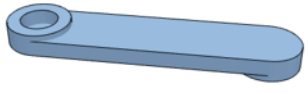
crank_mechanism_plate

1×



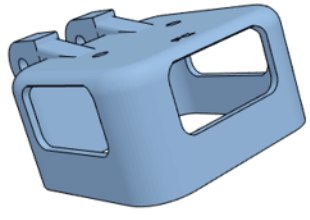
connecting_link_1

1×



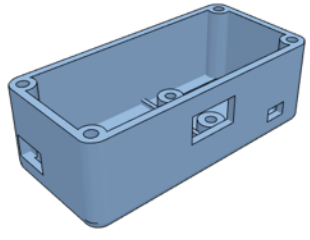
connecting_link_2

1×



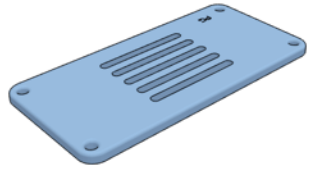
camera_mount

1×



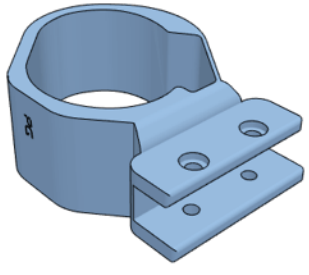
electronics_box

1×



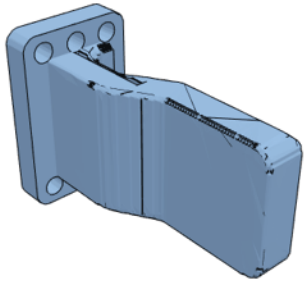
electronics_lid

1×



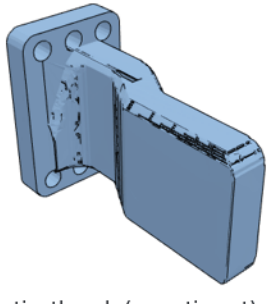
controller_support

1×



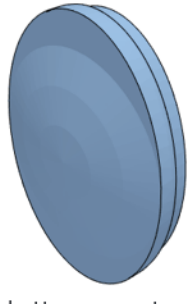
tip_index (your tip set)

1×



tip_thumb (your tip set)

1×



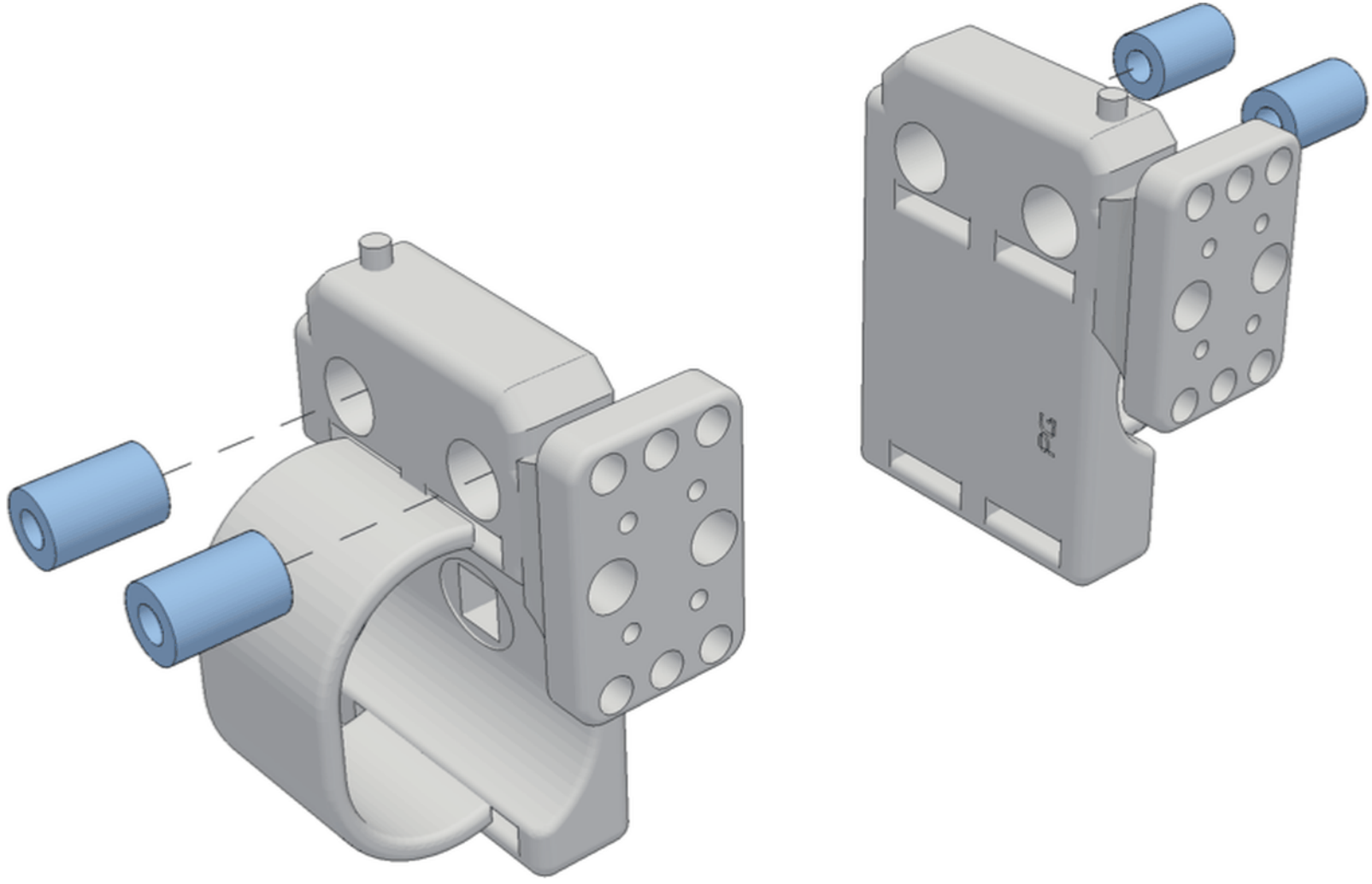
button_cover_tpu

1×

1

Linear bearings

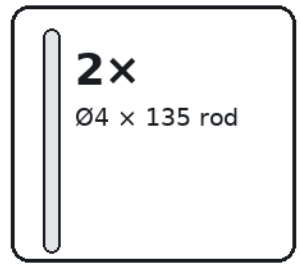
4×
LM4UU 4 × 8 × 12

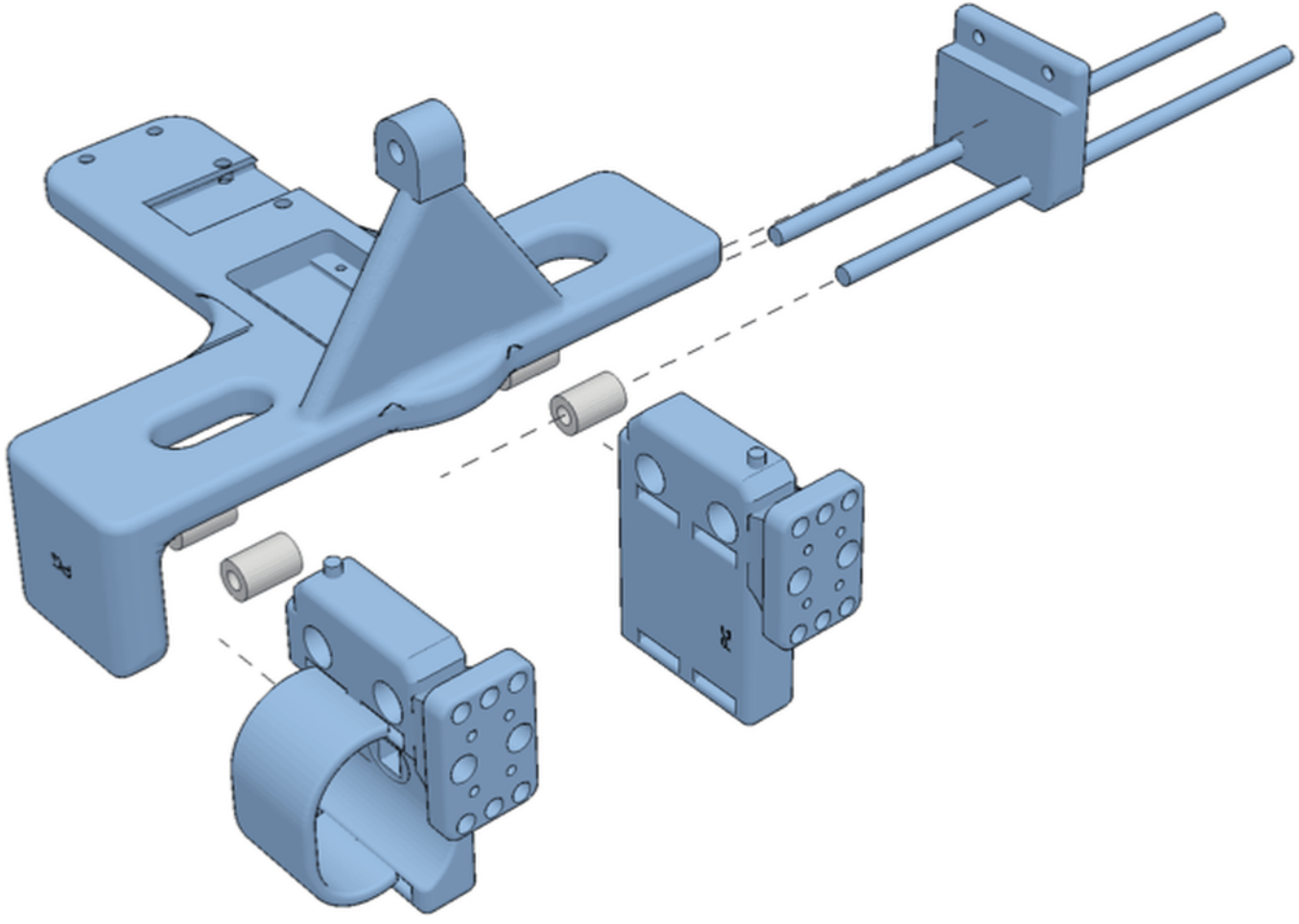


Press two LM4UU linear bearings into each finger link (index and thumb).

2

Rods and links



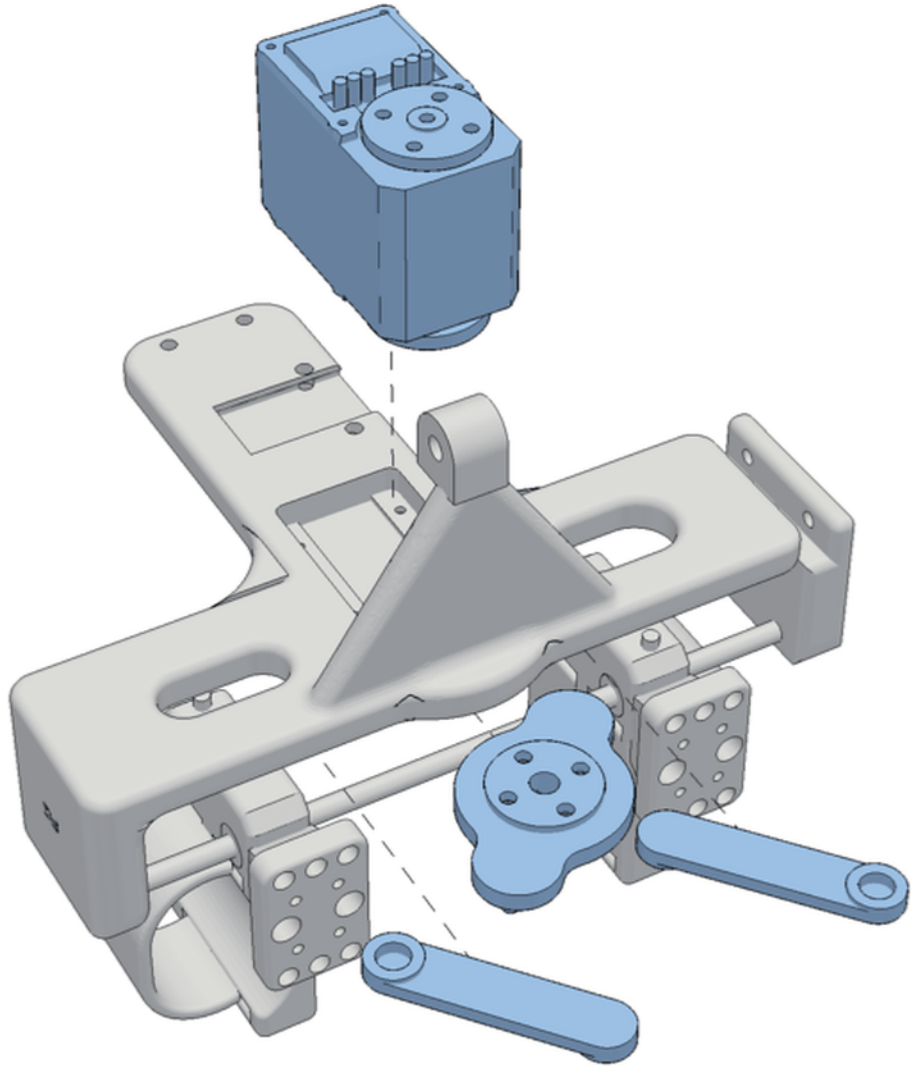


Hold both links in the main support and push the two Ø4 rods through the support and the links' bearings. Close the rod ends with the cover plate.

3

Encoder and linkage

1×	4×	4×	4×	2×
STS3215 servo	MR63ZZ 3 × 6 × 2.5	M3 × 12	M3 × 15	M3 × 20

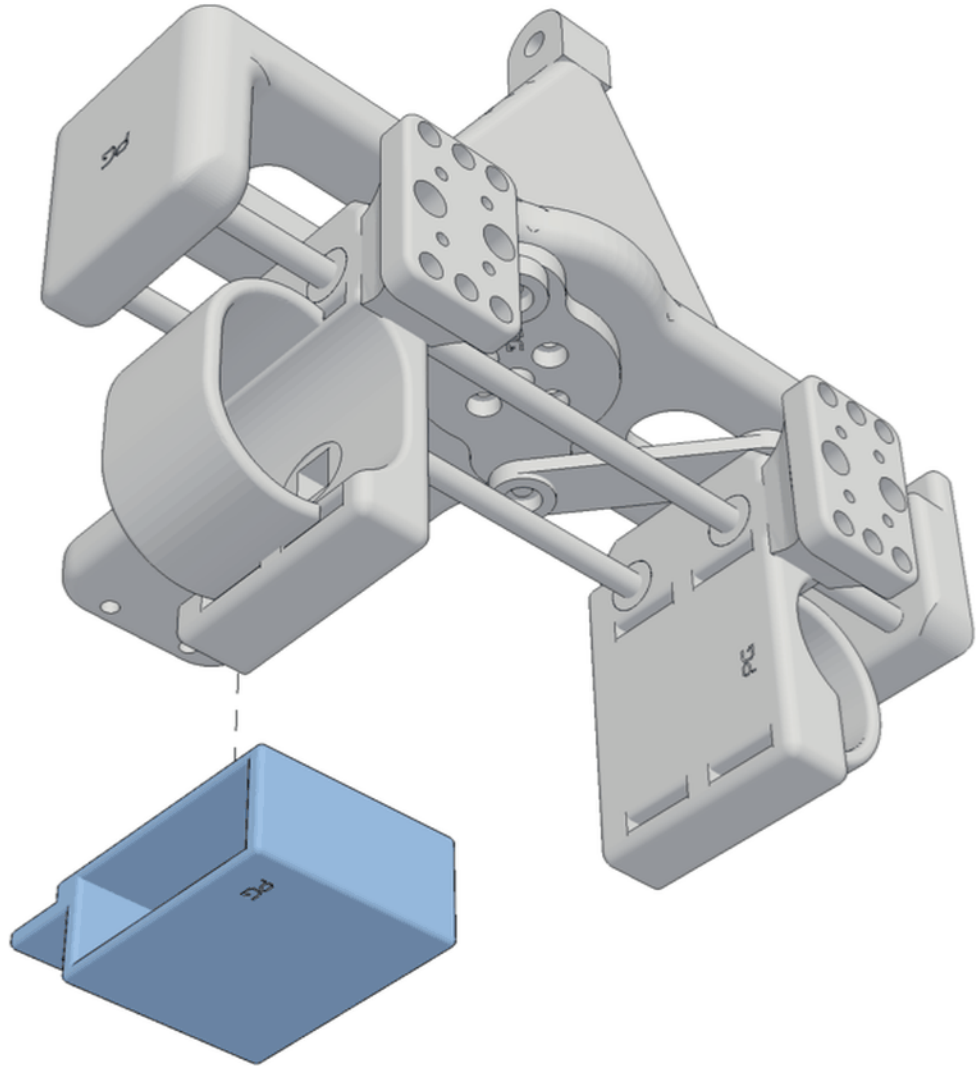


Seat the STS3215 (encoder, torque off) in the support. Fit the crank on its horn and join it to both links with the connecting links on MR63ZZ bearings. Screws: HandUMI's mechanism set.

4

Hand support

4×
hook-and-loop strap

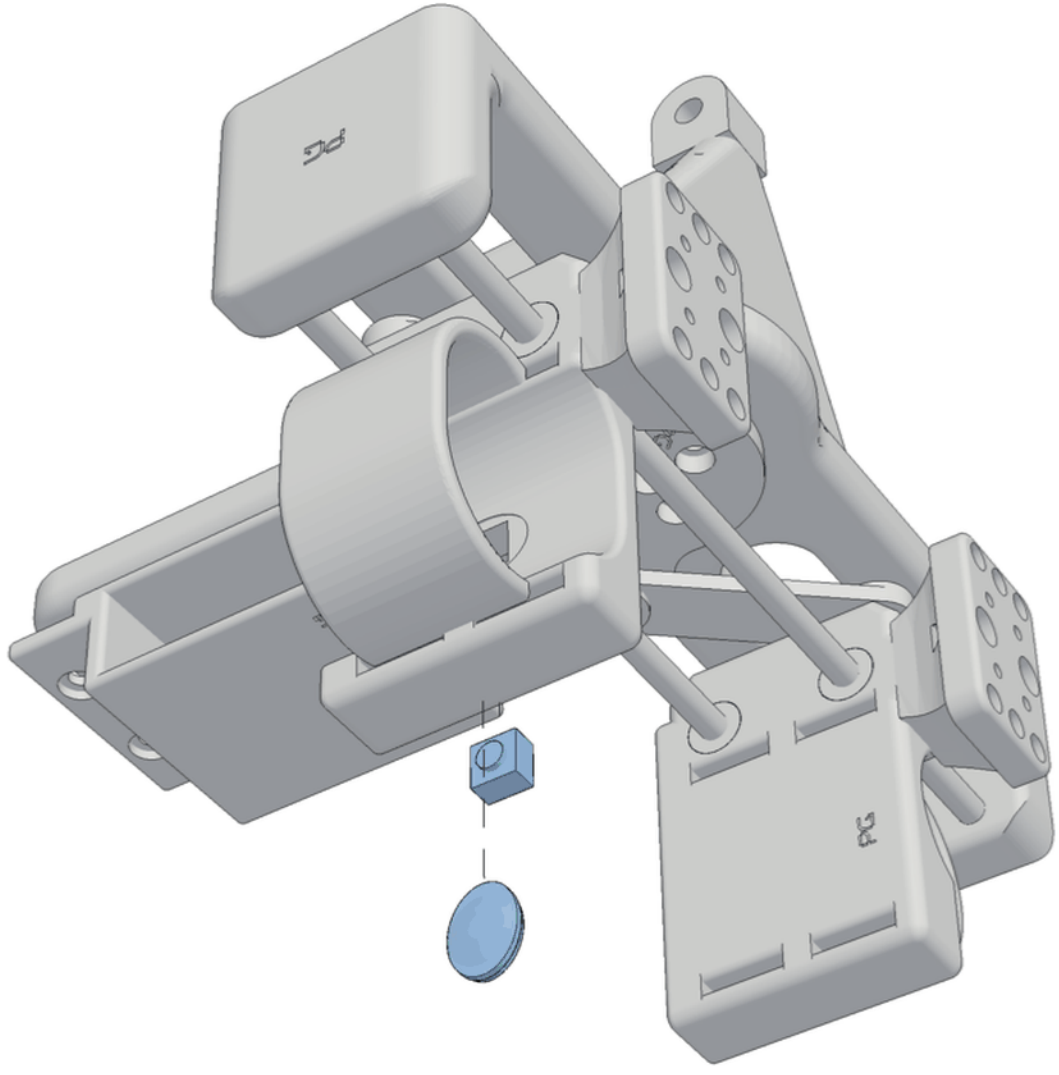


Screw the hand support under the main support and thread the hook-and-loop straps through it.

5

Record button

1×
6 × 6 tactile switch + wire

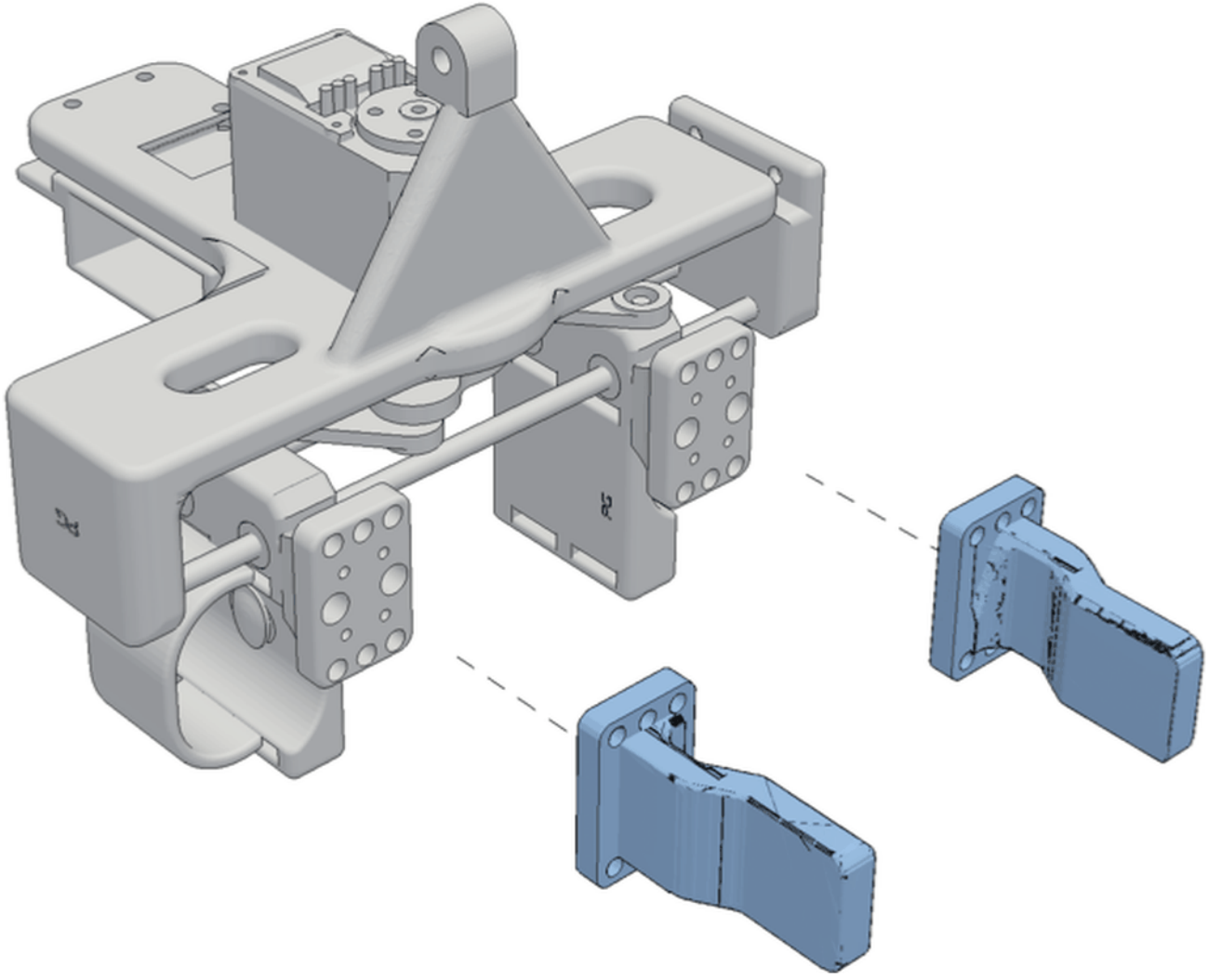


Lay the switch's wire in the index link's channel, press the switch in and push the TPU cover over it.

6

Tips

4×
M2 screws + nuts

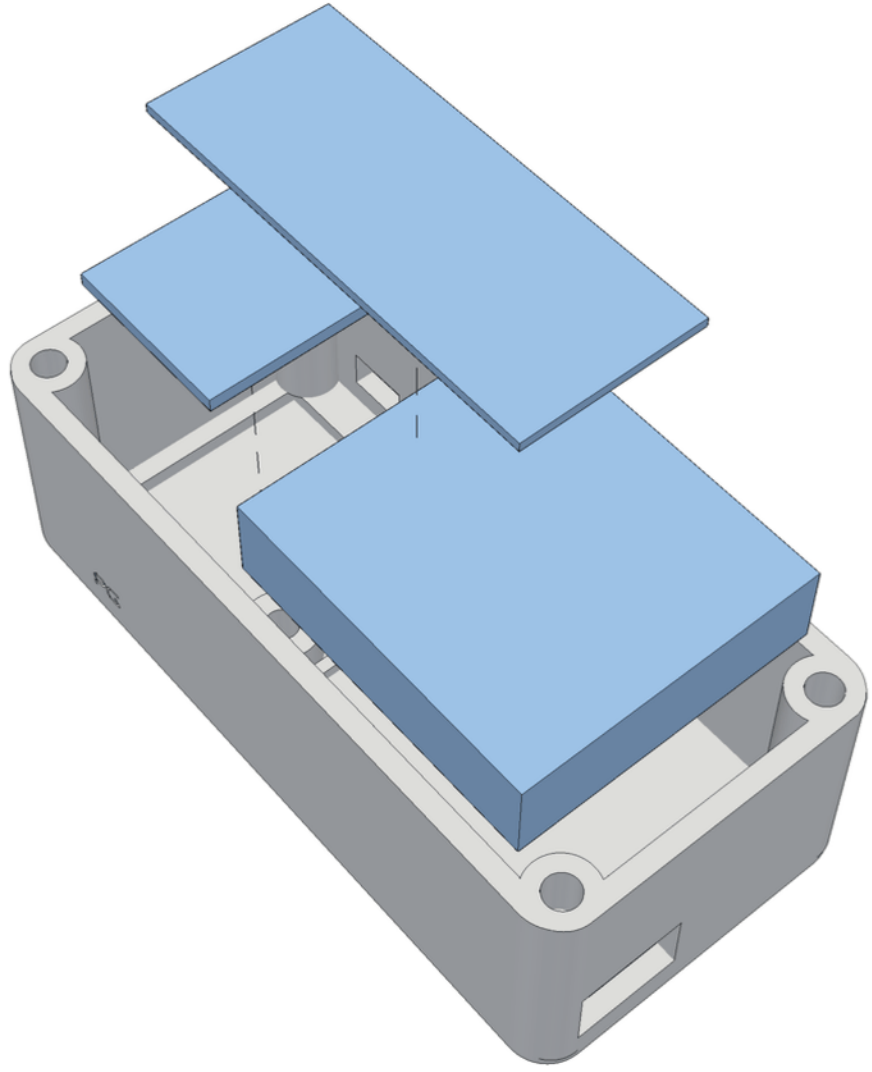


Screw the tip set for your robot (here: Robotiq Hand-E) onto the two links.

7

Electronics

4×
M2 screws (Pico)

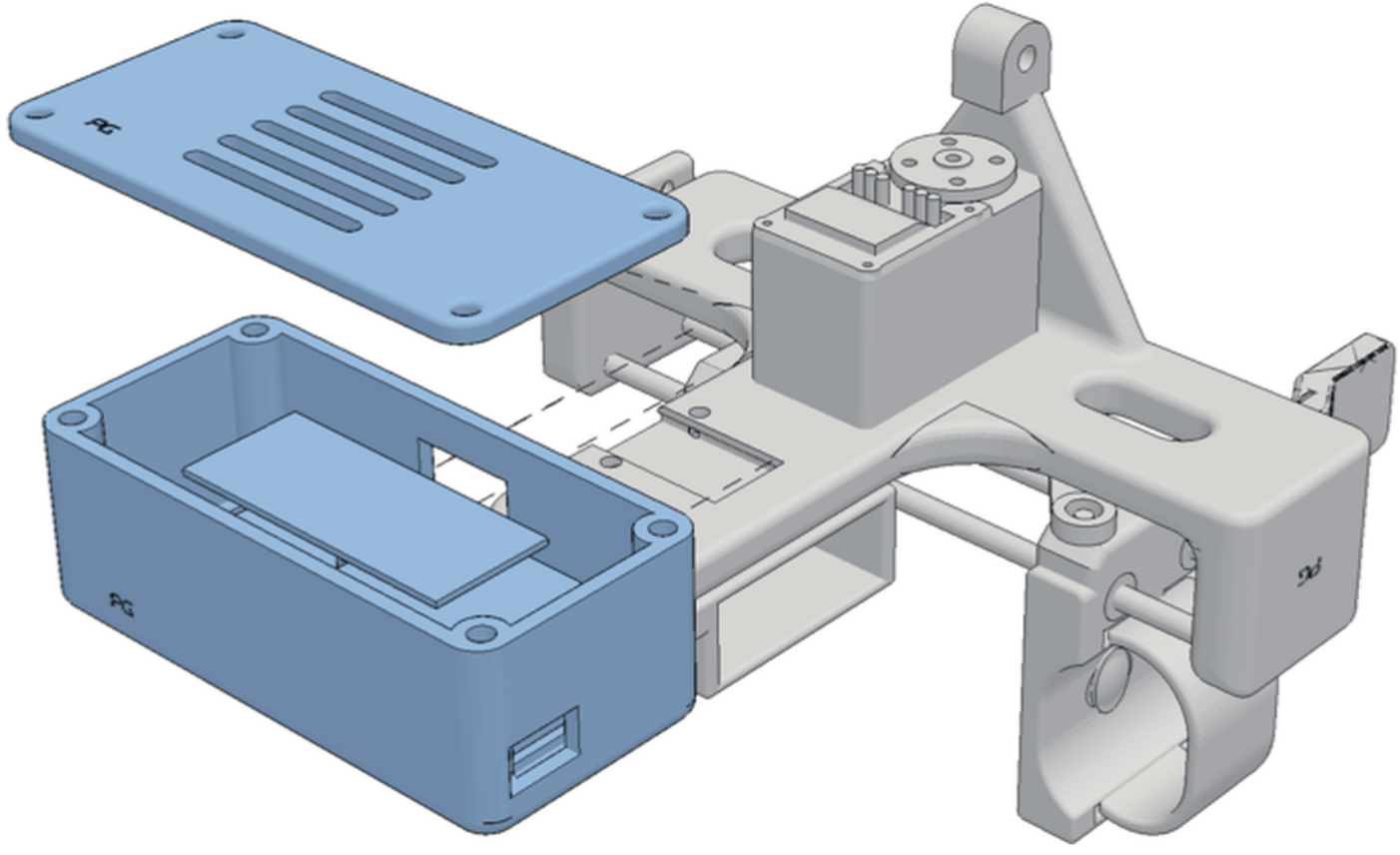


Fit the Pico 2, the IMU and the USB 3 hub in the electronics box and wire them as in docs/electronics.md.

8

Electronics box

4×	4×	8×
M3 × 8	M3 × 6 (lid)	M3 heat-set insert



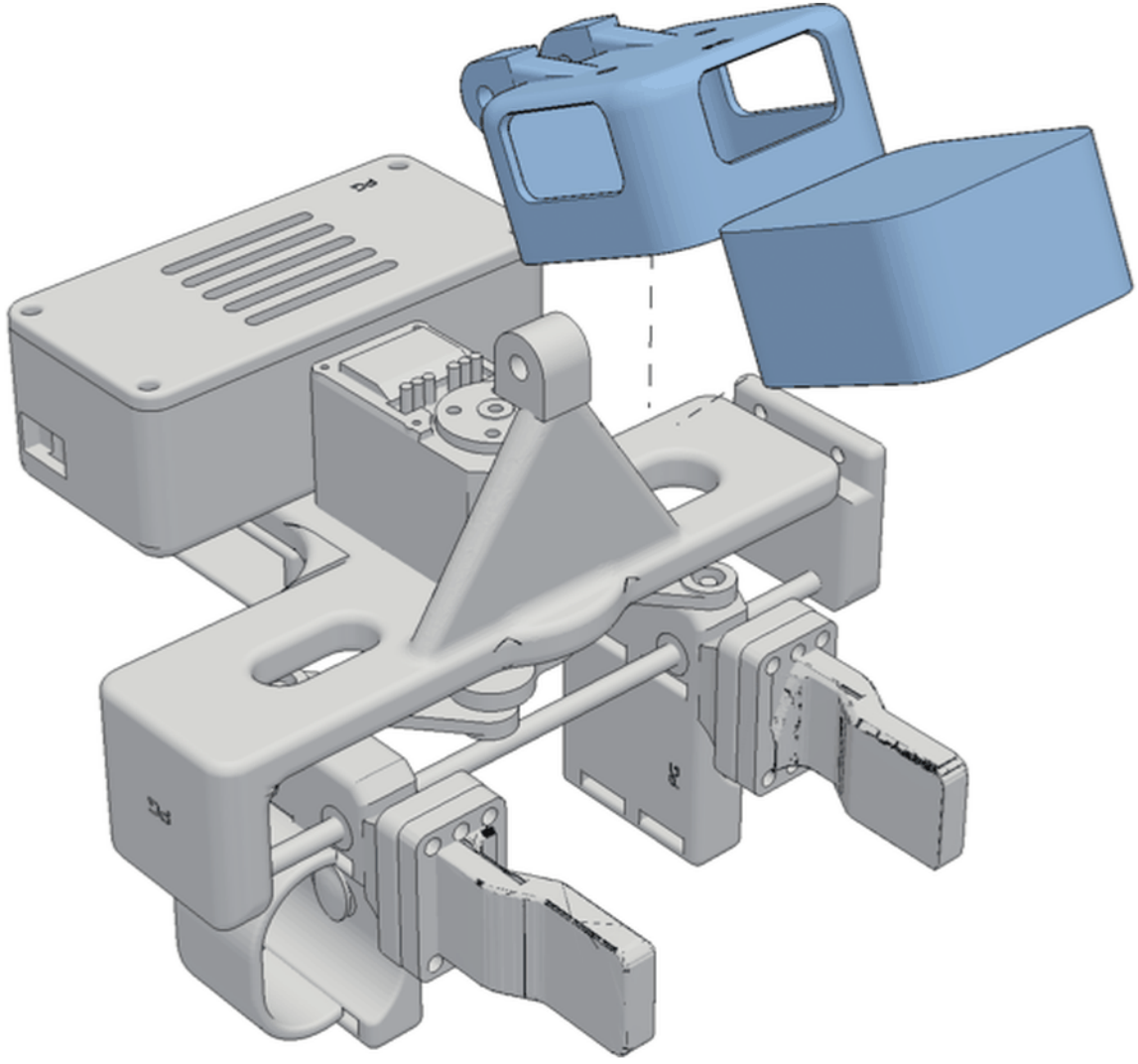
Screw the box to the support's arm (4 × M3 × 8), then close the lid (4 × M3 × 6 into inserts).

9

Camera

2× M3 × 5

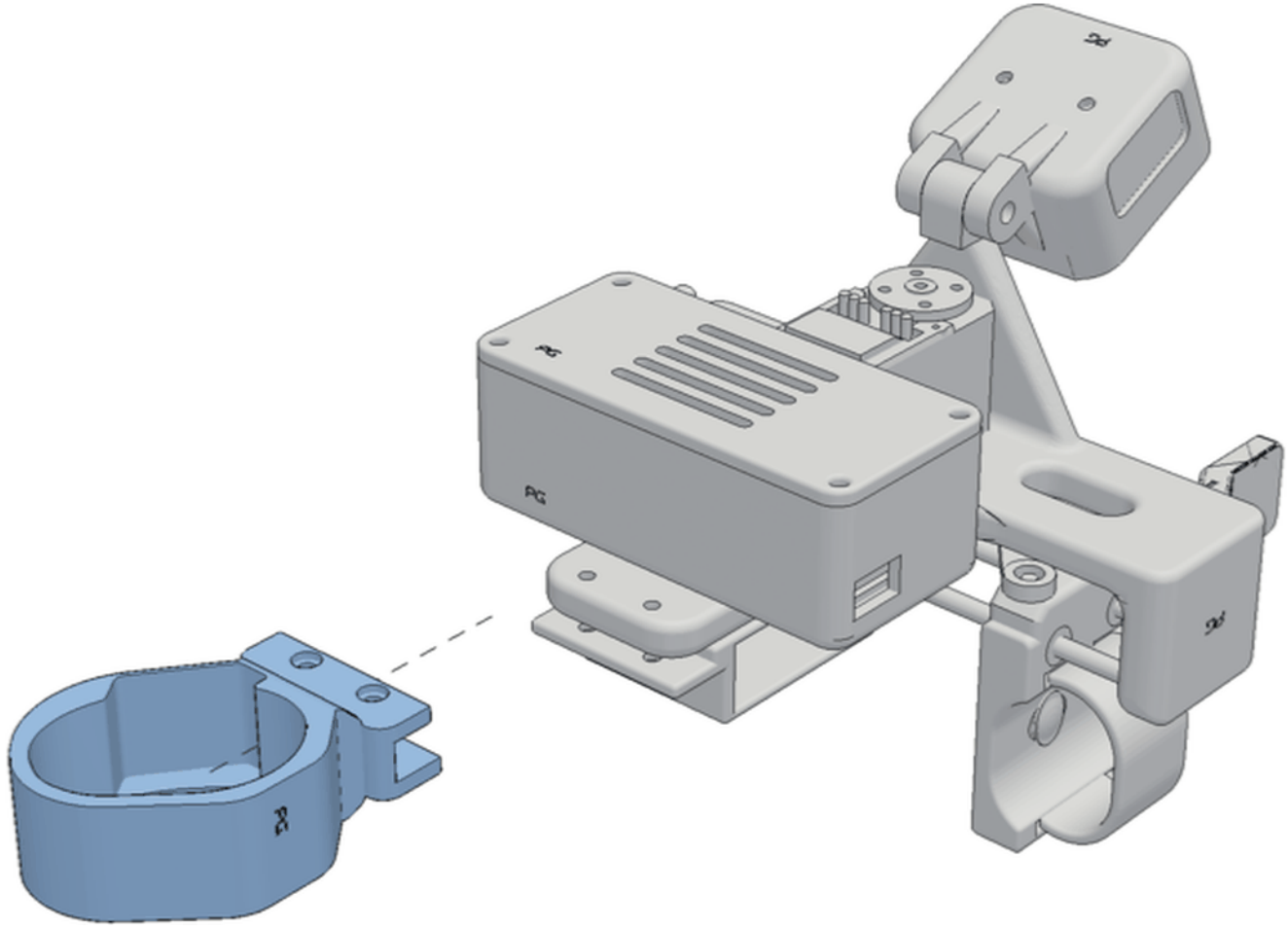
1× M4 × 30 + nyloc nut



Screw the D405 into its cup (2 × M3 × 5 into its rear holes). Hang the cup on the hinge with the M4 × 30 bolt and nyloc nut, set the tilt and tighten.

10

Controller support



Bolt the controller support to the back of the main support and push the Quest controller into its ring.